Microeconomics

Arseniy Stolyarov
PhD in Economics
University of Wisconsin–Madison

Artificial Intelligence and Financial Technologies
New Economic School · 2026/27

Economics is bigger than this course

One shared question, different lenses: **what shapes wages?**

CLASSICAL

Value and distribution
Land rent; wages and profits
Surplus and accumulation

Pay: shares of total output.

MARXIAN

Labour and surplus value
Exploitation; accumulation
Class and ownership

Pay: class and bargaining.

NEOCLASSICAL

Consumer and firm
Risk; intertemporal choice
Topkis: monotone statics

General equilibrium
Games and information

Pay: productivity and scarcity.

AUSTRIAN

Market process
Dispersed knowledge
Entrepreneurship

Pay: discovery and coordination.

KEYNESIAN

Effective demand
Post-Keynesian uncertainty
Financial instability

Pay: demand and employment.

INSTITUTIONAL

Rules and organisations
Property rights
Transaction costs

Pay: institutions and bargaining.

BEHAVIOURAL

Bounded rationality
Prospect theory
Social preferences

Pay: norms and reference points.

EVOLUTIONARY

Routines and selection
Innovation
Path dependence

Pay: changing technologies.

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→ An Amazon pricing case: mechanism, evidence, counterfactual

The course at a glance

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Final grade $= 0.20\,HW + 0.10\,Project + 0.70\,Exam$.

Each homework has two parts

1. **A mock exam.** Exam-level difficulty and length. Try it honestly in one **120-minute sitting**. The TA will mark it as a real exam so that you can assess your readiness. Errors in this part do not reduce your final course grade.

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Discuss approaches, but submit your own reasoning. If you use generative AI, disclose the tool and its role and be ready to explain every step.

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Topic after lecture 2; mechanism after lecture 4; final essay after lecture 6, before lecture 7. Presentations and discussion at seminar 7.

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The date, detailed format and permitted materials will be announced in advance.

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- Ask during class when a definition, assumption or argument is unclear.

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	Lecture	Seminar / milestone
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7	Intertemporal choice and finance	Saving, borrowing; project presentations; HW 3

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FTC: complaint and alleged anti-discounting practices

→ Back to why microeconomics

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Part II: useful assumptions

- Continuity and indifference curves
- Monotonicity, LNS and convexity
- Separability
- Quasilinearity

To model a choice, we need to know...

What do people **want**?

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What **can** they do?

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Start with what they want.

What is the primitive?

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- Quality, terms, timing and relevant context belong in the definition of an alternative.

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It is a statement about a pair of alternatives, not a numerical score or an observed purchase. Some introductory textbooks also introduce $\sim$ or $\succ$ separately. Do we need them as additional primitives?

Can we define indifference using only $\succeq$?

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Both comparisons hold. This differs from incomparability, where neither comparison holds.

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Does “not $x \succ y$” imply “$y \succ x$”?

No: $x \sim y$ is already a counterexample. Without completeness, incomparability is another.

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This does not require selfishness, perfect information or easy computation.

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Suppose the shared dishes, prices and information are identical, and each observed choice is the **unique favourite**. Can one fixed ranking explain both?

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For relation 2, try $(3, 2, 1)$, $(2, 1, 3)$ and $(1, 3, 2)$.

Condorcet’s paradox: rational voters, cyclic majority

Each voter has a complete and transitive ranking. Compare alternatives by simple majority.

Condorcet's paradox: rational voters, cyclic majority

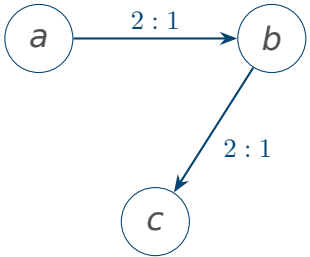
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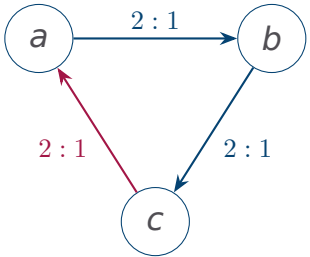
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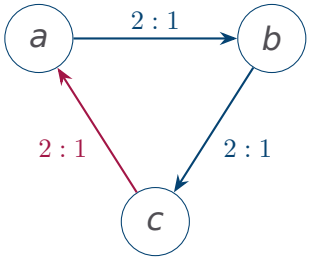
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Individual rationality need not produce a transitive collective ranking. No alternative beats every other alternative by majority.

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What must we observe besides a purchase before inferring anything about preferences?

Choice as the primitive

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A best element need not exist on an infinite menu. Nonemptiness is an additional requirement.

Consistency across menus: WARP

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NES

Weak Axiom of Revealed Preference

For any menus A, B and any $x, y \in A \cap B$,

$$\begin{aligned} & x \in C(A), \quad y \in C(B) \\ \Rightarrow \quad & x \in C(B), \quad y \in C(A). \end{aligned}$$

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- This formulation concerns the **whole correspondence**, including ties.

A two-menu inconsistency

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Menu $A = \{a, b, c\}$

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a is selected; b is available but not selected.

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Preferences and choice: an equivalence

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Let X be finite and C be nonempty on **every** nonempty subset of X.
Then C satisfies WARP if and only if there exists a complete, transitive relation $\succeq$ such that

$$C(A) = C(A; \succeq) \quad \text{for every nonempty } A \subseteq X.$$

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$\rightarrow$ Proof, if time permits $\rightarrow$ Preferences $\rightarrow$ WARP

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If we only know that an observed choice is optimal, universal indifference can rationalise any such observations. Additional structure matters.

Utility is a representation

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Definition

$u : X \to \mathbb{R}$ represents $\succeq$ if, for every $x, y \in X$,

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- Representation turns choice into an optimisation problem.

Representation on a finite set

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The ranking matters. The gaps between the labels do not.

Ordinal, not a happiness meter

Alternative	u	v
A	3	100
B	2	7
C	1	-20

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Both functions represent $A \succ B \succ C$.

Increasing transformations preserve the ranking

If f is strictly increasing on $u(X)$, then $v = f \circ u$ represents the same preferences:

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Examples: $u + 5$, $30u$, e^u. Squaring works on a nonnegative range, not in general. Why is a **weakly** increasing transformation not enough?

No interpersonal comparisons for free

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An individual ordinal representation does not, by itself, supply interpersonal units or a social welfare criterion.

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The non-representation proof uses uncountably many disjoint intervals; see the appendix.

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4. What changes when we move from finite to infinite choice sets?

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Lexicographic counterexample: $(1/n, 0) \succ_{\text{lex}} (0, 1)$ for all n, but $(0, 0) \prec_{\text{lex}} (0, 1)$ in the limit.

Indifference curves: equally good bundles

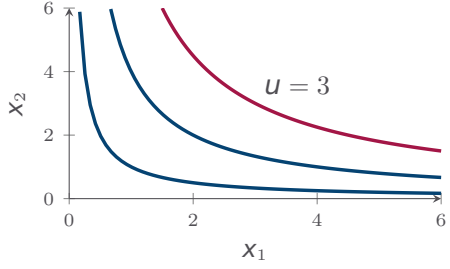
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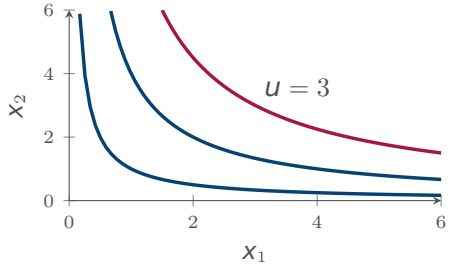
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These are level sets for the example $u(x) = \sqrt{x_1 x_2}$.
Different utility labels can describe the same indifference curves.

Monotonicity: fix the notation first

Write $x \geq y$ if $x_i \geq y_i$ for every i, and $x \gg y$ if $x_i > y_i$ for every i.

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- Neither is part of the definition of rationality.
- The assumption is inappropriate for a bad unless we redefine the commodity.

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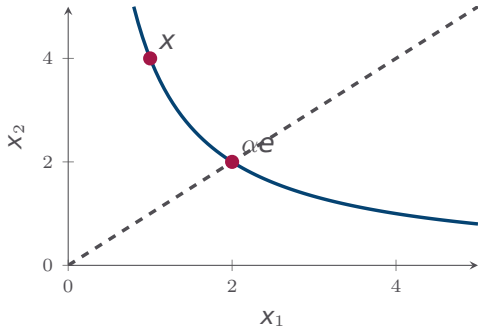
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Why only weakly better? With $u = \min\{x_1, x_2\}$, $(2, 3)$ and $(2, 2)$ are equally good. Strictness can disappear in a limit.

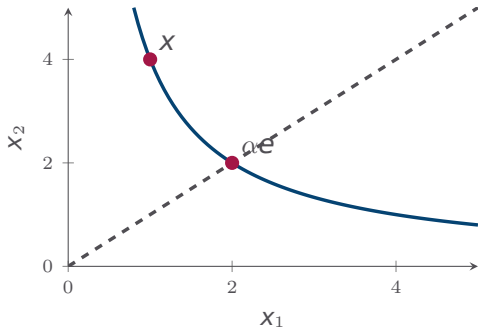
A constructive picture: the diagonal bundle

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Now add monotonicity to rationality and continuity. Each x is indifferent to a unique diagonal bundle αe, where $e = (1, \ldots, 1)$.

$$u(x) = \alpha \quad \text{if } x \sim \alpha e.$$

A useful way to **construct** a utility scale.

→ Proof under monotonicity

Local nonsatiation: a better bundle nearby

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- On $\mathbb{R}^L_+$: strong monotonicity $\Rightarrow$ monotonicity $\Rightarrow$ LNS.

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NES

Coffee and cake: could there be a bundle where every sufficiently small change makes you no better off?

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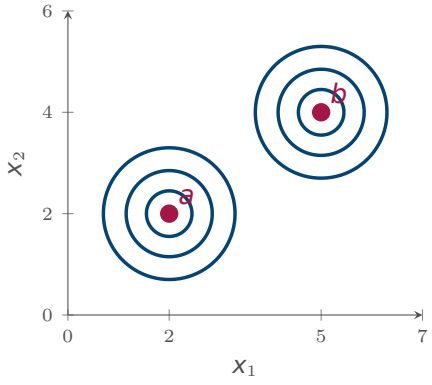
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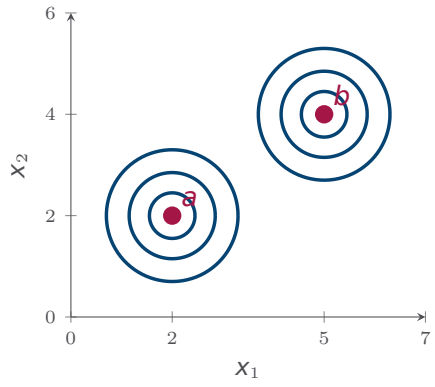
Example

$u(x_1, x_2) = x_1$ is locally nonsatiated but not strongly monotone.

Two bliss points



Two bliss points



Can these preferences be rational and continuous? Locally nonsatiated? Convex?

Convex preferences: upper contour sets

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Equivalent geometry

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If two distinct bundles are equally good, how does convexity rank their average?

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Half a sushi dinner and half a steak dinner, or some sushi and some steak over a month: is the same assumption equally plausible?

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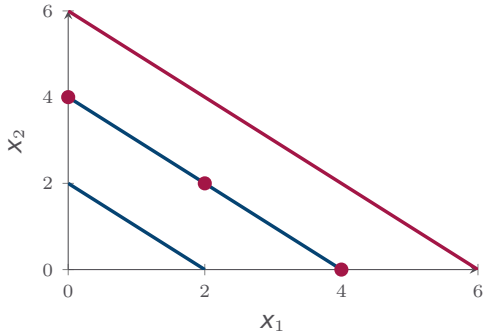
Same preferences, different curvature

$u(x) = x_1 + x_2$ is concave. $v(x) = e^{x_1+x_2}$ is not concave. They represent the same convex preferences.

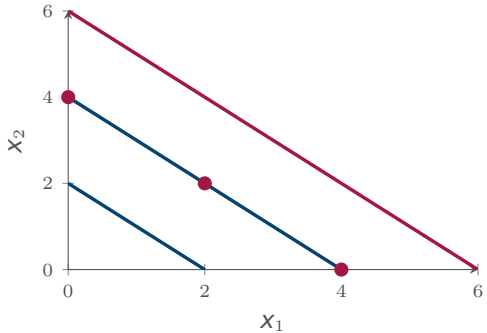
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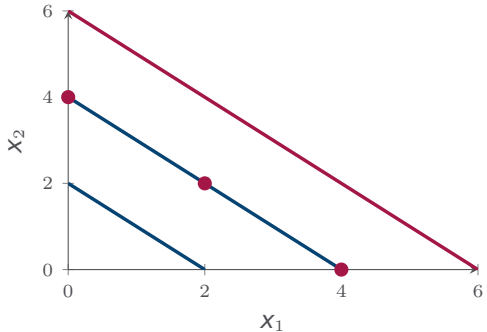
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One unit of either good contributes the same amount to the ranking.

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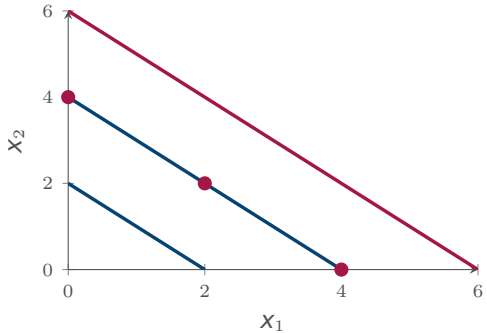


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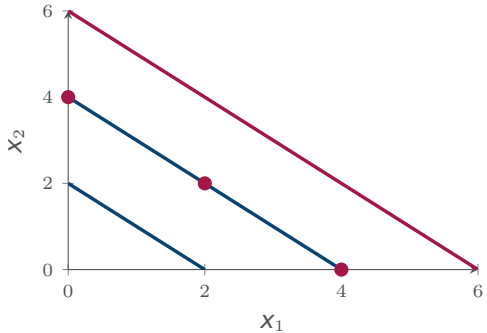


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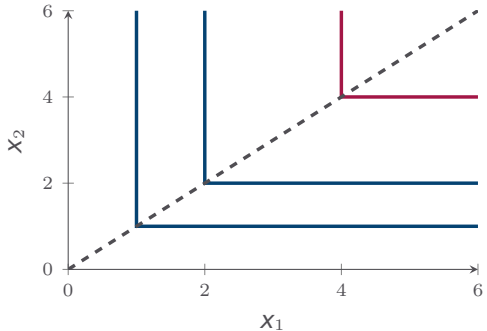
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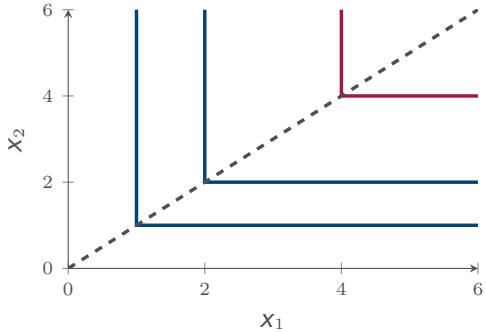
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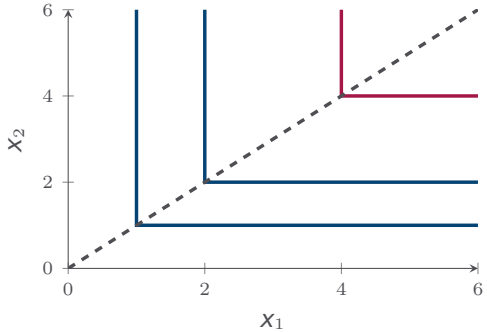
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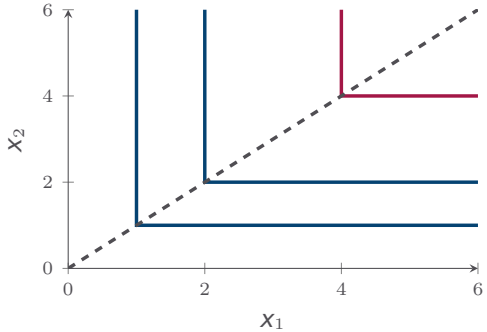


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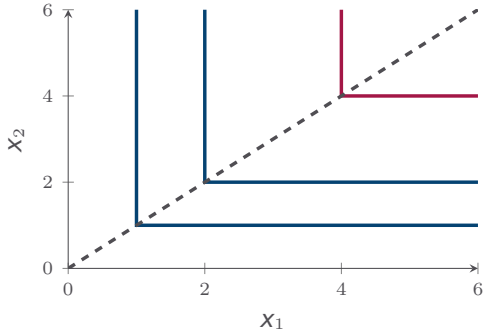


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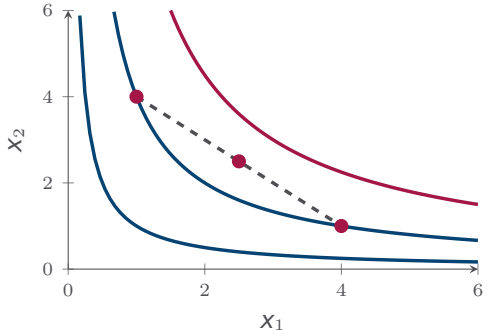
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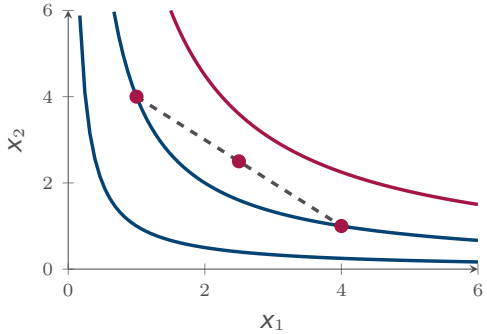
Cobb–Douglas: mixing can help

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$$u(x) = \sqrt{x_1 x_2}.$$

$u(1,4) = u(4,1) = 2$, but

$$u(2.5, 2.5) = 2.5.$$

On $\mathbb{R}^2_{++}$: strongly monotone and strictly convex preferences.
On $\mathbb{R}^2_{+}$, axis bundles have zero utility; the strict claims fail at the boundary.

Separability: a ranking inside a larger bundle

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- The condition is not symmetric between Y and Z.

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At **any fixed** z, the first food bundle ranks higher. This does not fix the food budget. → Formal construction

Quasilinearity: a numeraire plus a valuation

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A convenient assumption with economic content: money compensation is independent of the starting numeraire level.

No income effects: mind the boundary

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Let $u(a, y) = a + 4\sqrt{y}$, with both prices equal to 1 and $a + y \leq m$.
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Thus $y^* = \min\{m, 4\}$. Demand for y is income-independent when the numeraire constraint is slack; income matters when that constraint binds.

What can they do? The budget set

For $p_1, p_2 > 0$ and income $m > 0$,

$$B(p, m) = \{x \in \mathbb{R}^2_+ : p_1 x_1 + p_2 x_2 \leq m\}.$$

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A higher p_1 at fixed p_2, m: a rotation inwards around the vertical intercept.

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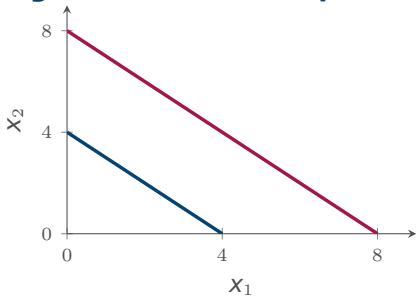
A higher p_1 at fixed p_2, m: a rotation inwards around the vertical intercept.

Preferences describe wants; the budget describes possibilities. Next lecture: solve the consumer's optimisation problem.

Budget geometry: income versus a price change

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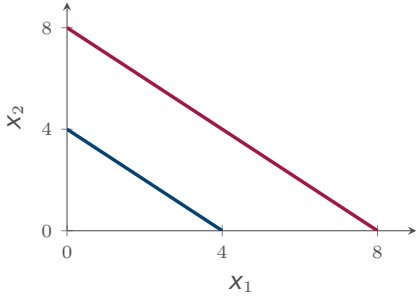
Higher income, fixed prices



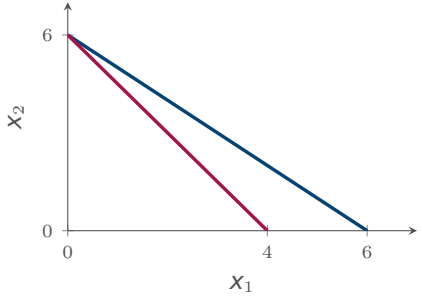
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NES

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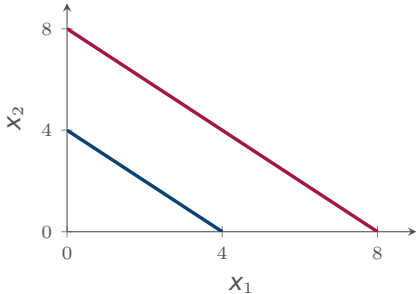


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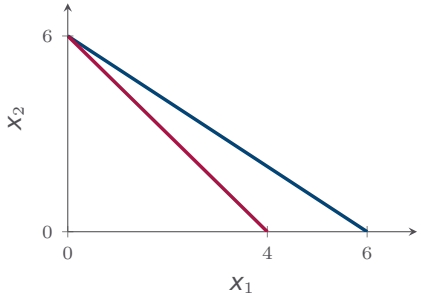


Budget geometry: income versus a price change

Higher income, fixed prices



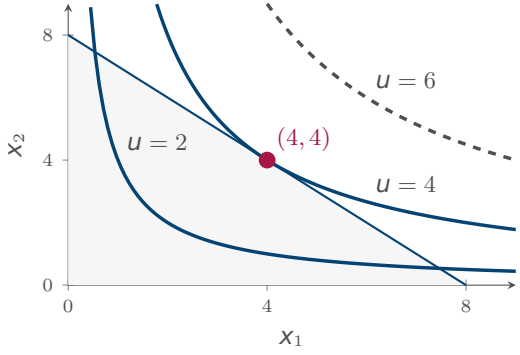
Higher p_1, fixed p_2, m



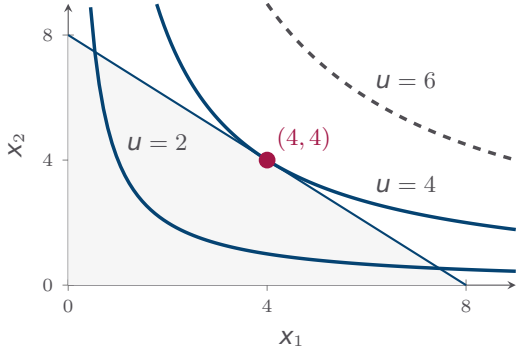
The feasible set expands in the first case and shrinks in the second. Neither change is itself a change in preferences.

Wants and possibilities in one picture

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Wants and possibilities in one picture



Possibilities:

$p_1 = p_2 = 1$, $m = 8$.

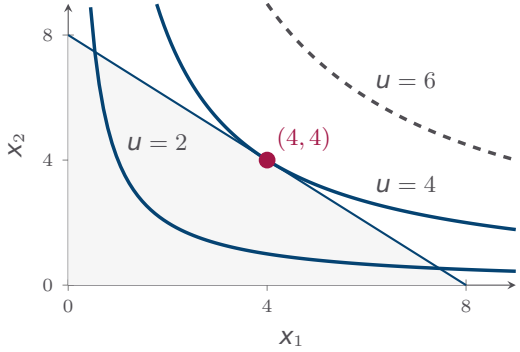
$x_1 + x_2 \leq 8$.

Wants:

$u(x) = \sqrt{x_1 x_2}$.

The best affordable bundle is $(4, 4)$, with utility 4.

Wants and possibilities in one picture



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Why is $u = 6$ unattainable? Why settle for $u = 2$?

Exit ticket

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2. Give rational preferences that are not strongly monotone.
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4. Does separability imply that spending on Y is independent of income?
5. What is the boundary caveat for quasilinear demand?

Use a definition, example or short argument for each answer.

What to keep, and what to read

NES

Main reading

Hal R. Varian, *Intermediate Microeconomics: A Modern Approach*, **9th edition** (2014).

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Chapter 1: *Preference and Choice*; Chapter 3: *Classical Demand Theory*, sections 3.B–3.C on preferences and utility.

Appendix: why rational preferences imply WARP

Suppose $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$.

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Appendix: why rational preferences imply WARP

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Suppose $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$.

1. Because $y \in A$, optimality of x gives $x \succeq y$.

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The argument uses transitivity and the full best-alternative interpretation of C. Finite menus ensure nonemptiness separately.

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Appendix: reconstructing the relation from choice

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- **Representation of choice:** if $x \in C(A)$, WARP with each $\{x, y\}$ yields $x \succeq y$. Conversely, if x beats every $y \in A$, choose $y \in C(A)$ and use WARP with $\{x, y\}$ to obtain $x \in C(A)$.

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Domain: all nonempty subsets of a finite X. Repeated alternatives in a triple give trivial cases.

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Appendix: limited data can hide a cycle

NES

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- No complete, transitive preference relation generates these full chosen sets.

Passing WARP on an arbitrary observed domain does not imply rationalisability on the full domain.

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Appendix: why lexicographic utility does not exist

Suppose a real-valued u represents lexicographic preferences on $[0,1]^2$.

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1. For each $t \in [0,1]$, $u(t,1) > u(t,0)$.
2. Thus $I_t = (u(t,0), u(t,1))$ is a nonempty open interval.

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There cannot be an injection from uncountably many $t \in [0,1]$ into the countable rationals. Contradiction.

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Appendix: construction under monotonicity

Assume rationality, continuity and monotonicity on $\mathbb{R}^L_+$; let $e = (1, \ldots, 1)$.

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1. Continuity and monotonicity imply $x \geq y \Rightarrow x \succeq y$.
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4. Transitivity and diagonal monotonicity give $x \succeq y \iff u(x) \geq u(y)$.
5. If $x^n \to x$, the coefficients $u(x^n)$ are bounded. Every convergent subsequence has limit $u(x)$ by continuity and uniqueness. Therefore $u(x^n) \to u(x)$.

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Appendix: separability in one construction

Fix z_0 and let $v(y) = u(y, z_0)$. For $t \in v(Y)$, pick $\hat{y}(t)$ with $v(\hat{y}(t)) = t$, and define

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- If $t > t'$, strict preference at z_0 carries to every z, so $U(t, z) > U(t', z)$.

The constructed aggregator has domain $v(Y) \times Z$.

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Appendix: primitives behind quasilinearity

On $\mathbb{R}_+ \times Y$, assume rationality, a worst $\bar{y}$, and:

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1. $(a, \bar{y}) \succeq (a', \bar{y}) \iff a \geq a'$.

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Transitivity and condition 1 then imply

$$(a, y) \succeq (a', y') \iff a + v(y) \geq a' + v(y').$$

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Transitivity and condition 1 then imply

$$(a, y) \succeq (a', y') \iff a + v(y) \geq a' + v(y').$$

So $u(a, y) = a + v(y)$ represents preferences.

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Appendix: LNS and budget exhaustion

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LNS implies budget exhaustion at an existing optimum; it does not establish existence.

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Appendix: increasing transformations and risk

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$$\mathbb{E}[A] = 2 > 1.5 = \mathbb{E}[B].$$

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$$\mathbb{E}[e^A] = e^2 \approx 7.39 < \frac{1 + e^3}{2} \approx 10.54 = \mathbb{E}[e^B].$$

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Appendix: increasing transformations and risk

Compare $A = 2$ for sure with $B = 0$ or 3, each with probability $1/2$.

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After the strictly increasing transformation $f(t) = e^t$,

$$\mathbb{E}[e^A] = e^2 \approx 7.39 < \frac{1 + e^3}{2} \approx 10.54 = \mathbb{E}[e^B].$$

The ranking of individual outcomes is unchanged, but the ranking of lotteries by expected transformed numbers changes. Ordinal deterministic utility and expected utility require different care.

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Appendix: answers to the relation exercise

1. Sum of coordinates: complete and transitive, by the ordering of real numbers.

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Appendix: answers to the relation exercise

NES

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2. At least two of three coordinates: complete, but not transitive. Let $a = (3, 2, 1)$, $b = (2, 1, 3)$, $c = (1, 3, 2)$. Then $a \succ b$, $b \succ c$, $c \succ a$.

→ Back to the exercise

Appendix: answers to the relation exercise

NES

1. Sum of coordinates: complete and transitive, by the ordering of real numbers.
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3. Componentwise comparison: transitive, but incomplete. $(2, 1)$ and $(1, 2)$ are incomparable.

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